THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The line $4x + y = 9$ is parallel to the line $kx + 5y = 11$. The value of k is: 1

(a) 21 **(b)** 20 **(c)** 19 **(d)** 22

2. A solid sphere of radius 5 cm is melted and recast into small solid spheres of radius 1 cm. The number of small spheres is: 1

(a) 150 **(b)** 120 **(c)** 130 **(d)** 125

3. The point on the y-axis equidistant from $(4, -1)$ and $(-4, 5)$ is: 1

(a) $(1, 2)$ **(b)** $(2, 0)$ **(c)** $(0, -2)$ **(d)** $(0, 2)$

4. A plan costs ₹100 plus ₹20 per unit; another costs ₹220 plus ₹10 per unit. They cost the same for ___ units. 1

(a) 11 **(b)** 14 **(c)** 12 **(d)** 13

5. If $x = 2 + \sqrt{3}$ with $2^2 - 3 = 1$, then $x + \frac{1}{x} =$ 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

6. The value of 101^3 is: 1

(a) 1030301 **(b)** 1030303 **(c)** 1030302 **(d)** 1030300

7. $ABCD$ is cyclic, $\angle ABD = 35^\circ$ and $\angle DBC = 25^\circ$. Then $\angle ABC =$ 1

(a) 60° **(b)** 50° **(c)** 80° **(d)** 70°

8. A wheel makes 100 revolutions to cover 220 m (take $\pi = \frac{22}{7}$). Its radius (cm) is: 1

(a) 36 **(b)** 34 **(c)** 37 **(d)** 35

9. A bag has 6 red balls and some blue balls. If the probability of drawing a blue ball is $\frac{2}{5}$, the number of blue balls is: 1

(a) 5 **(b)** 4 **(c)** 3 **(d)** 6

10. The sum of the first 12 terms of an AP is 510 and its first term is 4. The common difference is: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

11. The sum of 5 consecutive integers is 180. The largest of them is: 1

(a) 40 **(b)** 38 **(c)** 37 **(d)** 39

12. The average monthly salary of 22 employees is ₹66 thousand. An employee earning ₹25 thousand leaves and a new employee joins, and the average rises by ₹2 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 71 **(b)** 69 **(c)** 74 **(d)** 67

13. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 5702, the final r is: 1

(a) 2165 **(b)** 2075 **(c)** 2066 **(d)** 2084

14. A rhombus has side 51 cm and one diagonal 48 cm. The other diagonal is: 1

(a) 135 cm **(b)** 88 cm **(c)** 92 cm **(d)** 90 cm

15. The points $(-5, 19)$, $(-4, 18)$ and $(-3, k)$ are collinear. The value of k is: 1

(a) 16 **(b)** 18 **(c)** 19 **(d)** 17

16. A cuboid 54 cm $\times$ 12 cm $\times$ 6 cm is melted and recast into cubes of edge 3 cm. The number of cubes is: 1

(a) 146 **(b)** 150 **(c)** 148 **(d)** 144

17. **Assertion (A):** The converse of 'If a number is divisible by 6, then it is even' is false. 1
Reason (R): The number 4 is even but is not divisible by 6.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Class A has 20 students with average 60 marks and Class B has 30 students with average 70 marks; the combined average is 66. 1
Reason (R): The combined average equals the total of all marks divided by the total number of students.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The greatest common divisor of 48 and 18 is 6. 1
Reason (R): The gcd of two numbers is the largest number that divides both of them.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** The sum of the angles of a quadrilateral is 360°. 1
Reason (R): A diagonal divides a quadrilateral into two triangles, each with angle sum 180°.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. The line $kx + 3y = 27$ passes through the point $(1, 6)$. Find k, the slope and the y-intercept of the line. 2

22. A cuboid measures 6 cm × 18 cm × 27 cm. Find the length of its diagonal, its total surface area and its volume. 2

23. The end points of a diameter of a circle are $A(1, 1)$ and $B(13, 17)$. Find the centre and the radius. 2

24. If $p(x) = 3x + 4$, find $p(p(x))$ and its value at $x = 2$. 2

25. Find the smallest natural number n such that $\frac{n}{420}$ has a terminating decimal expansion. 2

26. Find the value of 103^3 using an identity. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. If $x = 2 + \sqrt{3}$, find $x^2 + \frac{1}{x^2}$. 3

28. Factorise $x^2 - 5x + 4$ and use it to find the value when $x = 6$. 3

29. Parallel chords of lengths 10 cm and 24 cm in a circle of radius 13 cm. Find the distance between them in both possible positions. 3

30. A running track is made of a rectangle 60 m $\times$ 28 m with a semicircle (radius 14 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

31. In 100 throws of a die, the number of times a prime number appeared was 52 and a number greater than 4 appeared 31 times (these include 14 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

OR

Find the sum of all natural numbers between 10 and 200 that leave remainder 2 when divided by 5. 3

32. Prove: if 3 divides x and x divides y, then 3 divides y. Verify with $x = 12$ and $y = 60$. 3

OR

A final score combines Maths (score 80, weight 4), Science (score 78, weight 3), English (score 54, weight 8), Hindi (score 85, weight 2). Find the weighted average score. 3

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SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** A circular garden in a Chandigarh park has radius 13 m. A straight path, 24 m long, joins two points on its boundary (a chord of the circle). 5

(i) Find the half-length of the chord.

(ii) Find the perpendicular distance of the path from the centre.

(iii) Find the length of another chord which is at a distance of 12 m from the centre.

(iv) If the two chords lie on opposite sides of the centre and are parallel, find the distance between them.

34. **Case study.** A circular pond of radius 14 m in a village in Kerala is surrounded by a 7 m wide path. Take $\pi = \dfrac{22}{7}$. 5

(i) Find the circumference of the pond.

(ii) Find the area of the pond.

(iii) Find the radius and the area of the outer circle formed by the path.

(iv) Find the area of the path and the cost of paving it at Rs 40 per m^2.

35. A bag has 7 red, 8 blue and 12 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{4}$. How many green balls were removed? 5

36. A ball is dropped from 81 m and rebounds to 2/3 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. 5

OR

Prove that the square of any odd integer is 1 more than a multiple of 8. Verify it for 9. 5

37. A family spent ₹6800 in January and ₹7600 in February. Food took 20% of the January total and 17% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

OR

Find the gcd of 882, 490 and 630 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 5

38. In a parallelogram ABCD the diagonals meet at O with $OA = 60$ cm, $OB = 63$ cm and $AB = 87$ cm. Show that ABCD is a rhombus and find its perimeter. 5

OR

A taxi fare is ₹176 for 2 km and ₹476 for 14 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 27 km. 5

END OF QUESTION PAPER — Total: 80 Marks